

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019**Start to Finish Automation Achieve on Cloud with Build Channel:
By DevOps Method**V. Arulkumar^a, R.Lathammanju^b^a SSN College of Engineering, Kalavakkam, Chennai.^b Saveetha School of Engineering, Thandalam, Chennai.

Abstract

Innovations through IoTs urge software developing industry need to come up with new life cycle model instead than usual. Process of software developments should be go along with advancements life cycle. This brings the attention for native and adoptable software life cycle process that is required for current industry standards. Manufactures are integrating the methodologies of software development process in a single bundle as automation with support of cloud with DevOps. The operation engineers, development engineers and QAs participate together for building the software applications will spin into minutes instead of days. Automation is the ultimate demand for DevOps approach. This approach advents continual automation to the software life cycle is known as delivery. This continually performs the tests and analysis of code. The developers provision and build the software into their cloud environment. Every step in the continuous delivery can be repeated in a several time a day depending on how much new applications need to release. Automation in DevOps improving speed of development, precision, consistency and increases the number of delivery.

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1. Introduction

Automate everything is the key principle of DevOps. Automation implies, Continuous Combination, Continuous Testing, Continuous Deployment and Continuous Conveyance. With the disposal of manual advances and the presentation of repeatable forms come numerous advantages that make computerization an imperative aggressive advantage. Automation in DevOps can lessen time to advertise from months and weeks to days and hours. Automation is particularly essential for little associations, which need to use their restricted assets as productively as conceivable to contend with built up players. Hence, the Automation turned into an enormous gain and decreased parcel of human mistakes not withstanding sparing time and permits any regular man do this, in this manner expelling the Automation became a huge.

1.1. Overview of DevOps

‘DevOps’ this word is a combination of development and operations. This is a culture which promotes collaboration between software developers and testing team to build software into release faster through automated and repeatable method. It advents to manufactures to build fast and speedup the delivery process and services. This helps the service providers and their customer’s better relationship and make healthy business in software industry.



Fig.1. Overview of DevOps

DevOps ought to be utilized for vast conveyed applications, for example, internet business locales or applications facilitated on a cloud stage. It ought not to be utilized in a mission-basic application like bank, control and other delicate information locales. Such applications need severe access controls on the generation condition, a point by point change the board arrangement, get to control approach to the server farms. Before DevOps, the improvement and activity group worked in complete disengagement. Coding and task groups have their different courses of events and are not in sync bringing about additional postponements. So there is an interest to expand the rate of programming conveyance by business partners. To accomplish the destinations of DevOps, Automate everything is an unquestionable requirement.

This work mainly focusing on automating the DevOps procedure. It empowers associations to decrease the usage time of new administrations from months to minutes. It expands the profitability of business and IT groups. It spares costs on support and updates, and wipe out pointless capital consumption. It institutionalizes the procedures for simple replication and quicker conveyance. It further improves the quality, dependability and reusability of all framework segments. Since it centers on conveying esteem a lot prior in an undertaking lifecycle, DevOps can be viewed as a perfect way to deal with national and government IT anticipates, just as huge scale ventures for the private part. It quickens new administrations through constant improvement and operational adaptability, giving inventive and practical ways for conveying an incentive through better approaches for improvement and tasks. This paper flows in the following ways as chapter 2 reviewing literature works, chapter 3 focusing with system design for the developed system, chapter 4 focusing DevOps implementation and chapter 5 listing results, conclusion and opportunities in DevOps in near future.

2. Literature Survey

This chapter manages different investigations and results that have been distributed in the past pertinent to DevOps automation. DevOps means start to finish mechanization in programming advancement and conveyance. The author in [1] checking on DevOps incorporates the two universes of improvement and activities, utilizing robotized advancement, organization, and foundation observing. It's an authoritative move wherein, rather than disseminated silted gatherings performing capacities independently, cross-practical groups take a shot at persistent operational element conveyances. The authors in [2] evaluating the product advancement procedure models utilized in the IT business pursues a typical system used to structure, plan and control the way toward building up a data framework in an effective and beneficial way. The authors in [3] states the objective of DevOps isn't to take out mistakes/personal time, rather automate manual assignments and make repeatable procedures. By this procedure, DevOps help associations achieve a steady and dependable IT foundation. In this manner, one of the drivers of expanding enthusiasm for DevOps practices and instruments is the critical decrease of spontaneous vacation and the expense related with it. The author in [4] proposed that the DevOps is a lot of practices planned to diminish the time between submitting a change to a framework and the change being put into ordinary creation, while guaranteeing high caliber.

3. System Architecture

This chapter deals with overall architecture of the system and various tools and packages used for implementation.

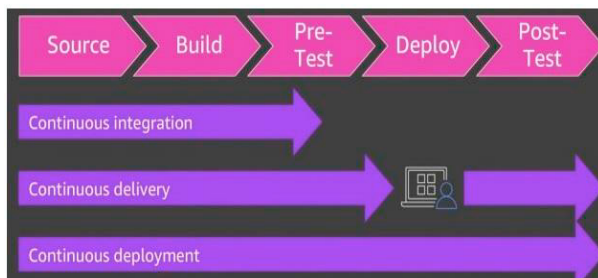


Fig 2: System Architecture

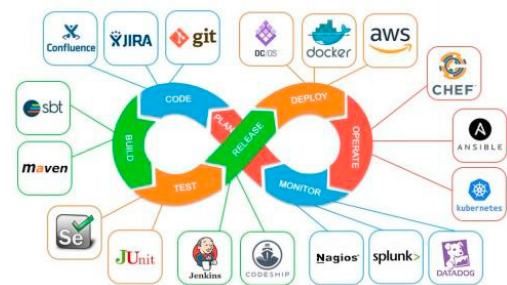


Fig 3: Tools Required

Source control in Fig 2 also known as version control allows developers to collaborate on code and track changes. It is one of the key tools used by successful DevOps teams for reducing development time and increasing the rate of successful deployments. Version control has an immediate impact on the quality and reliability of each deployment.

3.1 Tools and Environment Used

Amazon Web Services (AWS) started to offer IT services to the market in the form of web services, which is nowadays known as cloud computing as shown in fig 3. Maven is a project management and comprehension tool that provides developers a complete build lifecycle framework. Gulp is a cross-platform, streaming task runner that lets developers automate many development tasks. Protractor is an end-to-end test framework for Angular and AngularJS applications. Protractor can automatically execute the next step in your test, the moment the webpage finishes pending tasks, so you don't have to worry about waiting for your test and webpage to sync. It works as a solution integrator combining powerful technologies like Selenium, Jasmine, Web driver, etc.

4. Implementation

Our proposed methodology has the process flow as shown above fig 5. Our proposed system gathers the data from Code Commit. As soon as the data is fetched, it enters the build stage which is managed by the Code Build (a tool in AWS for managing the build process). The output from the build is then deployed to the specified environment in elastic beanstalk (fastest and simplest server to deploy our application on AWS). This corresponds to

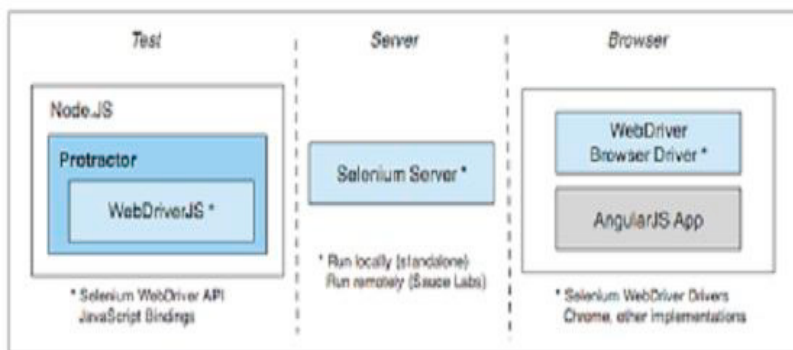


Fig 4: Testing Architecture

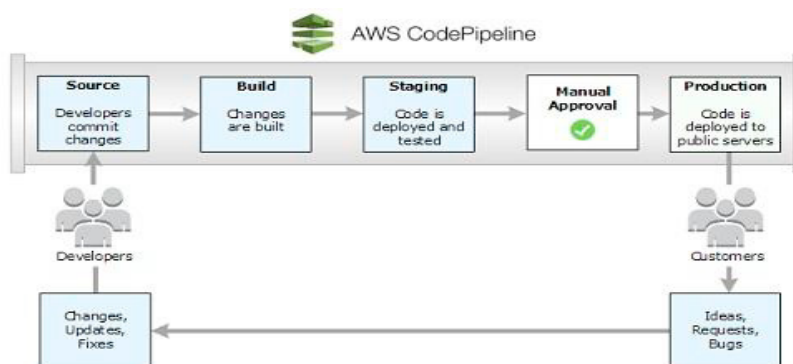


Fig 5: Proposed System

the backend automation. In a similar way the frontend automation is also performed. After continuous deployment, continuous testing is performed for checking the workflow of the application.

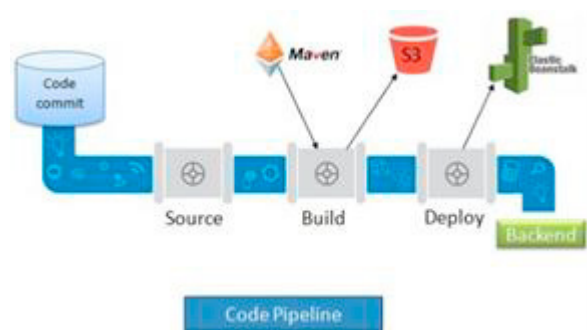


Fig 6: Backend Workflow

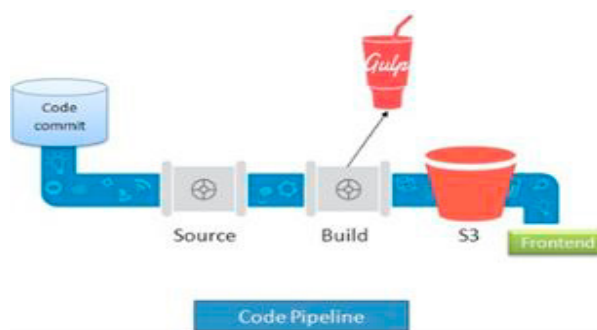


Fig 7: Frontend Workflow

Table 1. Steps followed for Back and Frontend Workflow.

Steps to be followed for Backend workflow	Steps to be followed for Backend workflow
Step1: For source: • Create the source code • Add a Build Spec File to the Source Code Step2: Create a build project with Code Build. Step3: Create an AWS Elastic Beanstalk Step4: Create the Pipeline and Deploy	Step1: Create the source code Step2: Create a build project with Code Build Step3: Create the Pipeline and deploy

4.1. Testing



Fig 8: Testing

Our work uses Protractor tool for testing. It is an end-to-end test framework for Angular and AngularJS applications.

Step 1: Setup

Install Protractor globally using npm, Update the server and start up the server.

Step 2: Write the test.

Step 3: Run the test.

4.2 Performance Metrics

1. Deployment frequency: Tracking how often we do deployments is a good DevOps metric.
2. Deployment time: Tracking how long it takes to do an actual deployment is another good metric.
3. Lead time: Determining how long it takes a change such as a bug fix or new feature to go from inception to production.
4. Automated tests pass %: To increase velocity, it is highly recommended that our team makes extensive usage of unit and functional testing.
5. Service availability: Application uptime is an important metric for every IT organization. Which creates uptime goals as high as 99.999%.

5. Result and Discussion

5.1 Automation

Our work functioning with the below outsets. Fig 9 gives valid Automation for the works we carried out. The fig 10: corresponds to the manual approval that was added to notify and get permission from the required team to proceed with the deployment. Fig 11. The build status shows whether the build was successful or not. It also includes the start time, end time and the output artifacts.

When we run the test, we will find that the browser is controlled by the automated test software where an end to end

testing takes place. This work has focused on automating the work of the development and operation team which in turn increases the productivity and quality of the product as it performs continuous integration, deployment and delivery at a faster rate than the manual method. The unique characteristics of the project are as follows: Respects the organization's culture, Automation, Business Benefits, Operational Benefits.

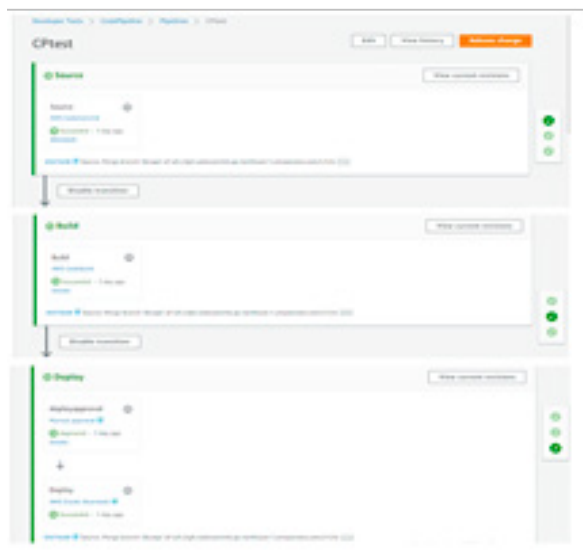


Fig 9: Backend Automation.

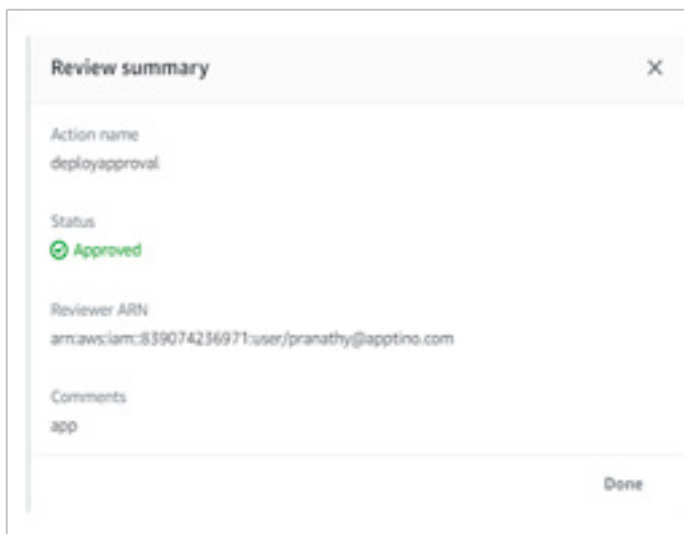


Fig 10: Review Summary

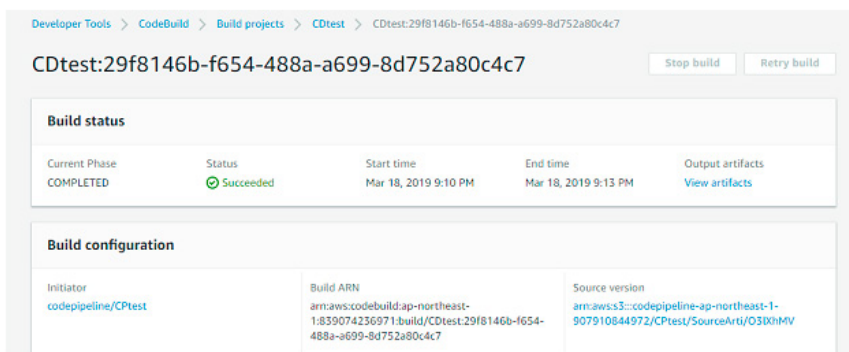


Fig 11: Start- End time Artifacts

